

Alex Byrne

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Education

Massachusetts Institute of Technology, Cambridge MA

August 2021 – May 2026

PhD in Physical Chemistry, GPA: 4.8, NSF Graduate Research Fellow

Thesis Title: Kinetic Modeling of Complex Carbon Chemistry in the Cold Interstellar Medium

Temple University College of Science and Technology, Philadelphia PA

August 2017 – May 2021

Bachelor of Science in Chemistry, GPA: 3.99

Graduated summa cum laude and with distinction in major

Skills

- **Programming:** Python, Git, SQL, Linux OS, Shell Scripting, Parallel Computing
- **Analysis & Visualization:** Kinetic Modeling, Jupyter, Pandas, NumPy, Dask, Matplotlib, Scipy
- **Machine Learning:** PyTorch, Scikit-learn, Cheminformatics (SMILES, SELFIES, RDKit), Unsupervised Learning (Clustering, Dimensionality Reduction, Vector Embedding), Supervised Learning
- **Communication & Collaboration:** Microsoft Office Suite, Google Office Suite, Slack, LaTeX, Scientific Writing, Public Speaking, Multidisciplinary Collaboration, French (Intermediate)
- **Certificates:** MIT Research Mentoring Certificate, American Chemical Society Degree Certification
- **Relevant Coursework:** Atmospheric Chemistry, Isotopes and Understanding the Universe, Modeling with Machine Learning, Machine Learning for Molecular Engineering, Kinetics of Chemical Reactions

Research Experience

Research Specialist, Massachusetts Institute of Technology

June 2026 – Present

- Spearheaded computational modeling and data management projects for a research community spanning more than 12 institutions and 5 countries.
- Improved performance of a kinetic model of hydrocarbon chemistry by a factor of 1000 with thorough updates and benchmarking.
- Devised and implemented standard operating procedures using Git workflows to ensure data quality, accessibility, and transparency.

Graduate Research Assistant, Massachusetts Institute of Technology

November 2021 – May 2026

Department of Chemistry, Advisors: Brett McGuire and Troy Van Voorhis

- Developed kinetic models, data analysis scripts, and computational workflows using Python and shell scripting to investigate low-temperature chemistry of hydrocarbon pollutants.
- Independently designed and executed sensitivity analyses of over 7000 model parameters, guiding future studies by identifying the 10 most influential sources of uncertainty.
- Developed a pipeline to visualize over 1000 molecules in two-dimensional vector space using web scraping, machine learning in PyTorch, and UMAP dimensionality reduction.
- Published 3 first-author journal articles peer-reviewed 3 publications in chemistry and physics.
- Mentored 2 undergraduate students and 9 graduate students on research and navigating higher education.

Undergraduate Researcher, Temple University

May 2018 – May 2021

Department of Chemistry, Advisor: Michael Zdilla

- Conducted organic syntheses and analyzed products via NMR, UV-vis spectroscopy, and X-ray diffraction.
- Performed quantum chemical calculations using the MoPro and Gaussian suites.

Teaching Experience

Teaching Assistant, Massachusetts Institute of Technology

August 2021 – May 2022

Principles of Chemical Science and Thermodynamics and Kinetics

- Led weekly recitations and office hours for reviewing and practicing class material.
- Provided grading and feedback on exams and problem sets.
- Planned and directed class-wide exam review sessions for more than 150 students.
- Arranged chemistry demonstrations applying concepts covered in class.

STEM Tutor, Temple University Student Success Center

August 2018 – May 2021

- Guided students in mastering complex concepts in chemistry, calculus, and physics.
- Supervised exam review workshops and encouraged student collaboration.
- Trained new tutors in identifying the needs of the student and applying the appropriate techniques.

Projects

BATMAN Chemical Model - <https://github.com/cixue/BATMAN>

- Led the design and management of an open-source chemical model for an international collaboration.
- Developed and implemented standard operating procedures for model changes using Git workflows.
- Incorporated and documented the addition of more than 600 new parameters over multiple branches.

Sensitivity Analysis of Gas-Phase Chemistry in TMC-1 - <https://doi.org/10.5281/zenodo.13257329>

- Performed multiple sensitivity analyses of model behavior to over 7000 input parameters.
- Identified critical sources of model uncertainty, guiding future efforts to improve data quality.
- Published an online repository containing data, analysis scripts, and plotting scripts with 864 downloads.

Select Peer-Reviewed Journal Articles

- **Byrne, A. N.** et al., "Modeling the Effect of C/O Ratio on Complex Carbon Chemistry in Cold Molecular Clouds", **2026**, *The Astrophysical Journal*, 998, 95
- Xue, C., **Byrne, A. N.**, et al., "The Molecular Inventory of TMC-1 with GOTHAM Observations", **2026**, *The Astrophysical Journal Supplement Series*, 281, 9
- Keller, T., **Byrne, A. N.**, Prakash, P., Zdilla, M., "Determination of Partial Atomic Charges from Single-Crystal X-Ray Diffraction by Refinement of Occupancy of Spherical Atoms (ROSA)", **2025**, *Journal of Applied Crystallography*, in review
- **Byrne, A. N.**, Xue, C., Van Voorhis, T., McGuire, B. A., "Sensitivity Analysis of Aromatic Chemistry to Gas-Phase Kinetics in a Dark Molecular Cloud Model", **2024**, *Physical Chemistry Chemical Physics*, 26, 26734
- Wenzel, G., Speak, T. H., Changala, P. B., Willis, R. H. J., Burkhardt, A. M., Zhang, S., Bergin, E. A., **Byrne, A. N.**, et al., "Detections of interstellar 2-cyanopyrene and 4-cyanopyrene in TMC-1", **2024**, *Nature Astronomy*, 9, 262
- Wenzel, G., Cooke, I. R., Changala, P. B., Bergin, E. A., Zhang, S., Burkhardt, A. M., **Byrne, A. N.**, et al., "Detection of interstellar 1-cyanopyrene: A four-ring polycyclic aromatic hydrocarbon", **2024**, *Science*, 386, 810
- Vaddypally, S., **Byrne, A. N.**, Goldsmith, C. F., Zdilla, M. J., Kiselev, V., "Metal-Free Reversible Double Cyclization of Cyanuric Diazide to an Asymmetric Bitetrazolate via Cleavage of the Six-Membered Aromatic Ring", **2024**, *Inorganic Chemistry*, 63, 2322
- **Byrne, A. N.**, Xue, C., Cooke, I. R., McCarthy, M. C., McGuire, B. A., "Astrochemical Modeling of Propargyl Radical Chemistry in TMC-1", **2023**, *The Astrophysical Journal*, 957, 88
- Fried, Z. T. P., Lee, K. L. K., **Byrne, A. N.**, McGuire, B. A., "Implementation of rare isotopologues into the machine learning of the chemical inventory of the solar-type protostellar source IRAS 16293-2422", **2023**, *Digital Discovery*, 2, 952

Select Conference Presentations

- **Byrne, A. N.** et al., "Analyzing Sensitivity of Molecular Cloud Chemistry to Gas-Phase Kinetics", American Association for the Advancement of Science National Meeting, **2026**, Poster Presentation
- **Byrne, A. N.** et al., "Modeling the Effect of C/O Ratio in Cold Molecular Clouds: Key Chemical Effects and Model Nonlinearities", International Symposium on Molecular Spectroscopy, **2025**, Oral Presentation
- **Byrne, A.N.**, Xue, C., Van Voorhis, T., McGuire, B. A., "Sensitivity Analysis of an Interstellar Dark Cloud Chemical Model and Application to Aromatic Chemistry", Gordon Research Seminar: Modeling and Simulations to Understand Chemical Systems, **2024**, Oral & Poster Presentations
- **Byrne, A.N.**, Xue, C., Van Voorhis, T., McGuire, B. A., "Sensitivity Analyses of a Dark Cloud Model and Application to Interstellar Aromatic Chemistry", International Symposium on Molecular Spectroscopy, **2024**, Oral Presentation
- **Byrne, A.N.**, Xue, C., Van Voorhis, T., McGuire, B. A., "Sensitivity Analysis of a Dark Molecular Cloud Model and Application to Aromatic Chemistry", QuantumGrain Workshop: Emerging Horizons in the Chemistry of the Universe, **2024**, Oral Presentation
- **Byrne, A. N.**, Xue, C., Cooke, I. R., McCarthy, M. C., McGuire, B. A., "Astrochemical Modeling of Propargyl Radical Chemistry in Cold Molecular Clouds", IAUS 383: Astrochemistry VIII: From the First Galaxies to the Formation of Habitable Worlds (Kavli - IAU), **2023**, Poster Presentation
- **Byrne, A. N.**, Xue, C., Cooke, I. R., McCarthy, M. C., McGuire, B. A., "Astrochemical Modeling of Propargyl Radical Chemistry in Cold Molecular Clouds", International Symposium on Molecular Spectroscopy, **2023**, Oral Presentation

Awards and Honors

- NSF Graduate Research Fellow
- ACS Undergraduate Award in Inorganic Chemistry – Awarded to one student each year for demonstrated excellence in inorganic chemistry at undergraduate level
- John A. Poole Memorial Scholarship – Awarded for academic excellence
- Temple University Provost's Scholarship

Volunteering and Outreach

Committee on the Future of Graduate Education

December 2025 – May 2026

Developed a policy memo detailing issues within the current structure of graduate education and possible solutions, presenting these findings to congressional staffers and science policy experts.

Organizing Member, Scientist Network for Advancing Policy

May 2025 – Present

Organize congressional visits, write and edit for a science policy blog, and establish a science policy podcast as a member of a nationwide, grassroots organization of early-career researchers.

GSC EAB Public Outreach Co-Chair, Massachusetts Institute of Technology

April 2025 – May 2026

Coordinated with other internal and external policy-related organizations to develop science advocacy and communication initiatives, such as op-ed writing and high school outreach.

Astrochem Coffee Podcaster, Public Podcast

March 2024 – Present

Curate content for the Astrochem Coffee podcast by reading and summarizing recent papers of relevance to astrochemistry and interviewing researchers at the graduate level and beyond.

CGSC Quality of Life Representative, Massachusetts Institute of Technology

November 2023 – May 2026

Discussed issues pertaining to quality of life in the chemistry department with graduate students, postdoctoral scholars, administrative staff, and faculty as a representative of the Chemistry Graduate Student Committee.

CAMP Mentor, Massachusetts Institute of Technology

October 2022 – November 2024

Guided prospective students from underrepresented backgrounds through MIT chemistry graduate program application and provided feedback on professional statements.